Software Requirements Specification
for
MEETSPACE Smart Conference Room

Version 1.0 October 2026

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Course Software Requirements Analysis and Specification

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Revision History

Version	Date	Description
1.0	23 October 2026	Initial SRS based on supplied template, reference SRS, project presentation, diagrams and current application source.

Document Approval

Prepared for stakeholder review. Approval has not yet been recorded.

Name	Role	Signature	Date
To be completed	Stakeholder		

1 Introduction

This document specifies MEETSPACE, a browser based meeting room reservation system with a main member website and a separate Walk-in kiosk. It defines required behavior, access rights, validation, interfaces and stored data for Release 1. Requirements reflect the current application source inspected on 23 October 2026. Earlier analysis diagrams are retained as references with differences identified in Appendix B.

1.1 Business Problem

Staff lack complete information about meeting room equipment and may see a room as available although someone is already using it. This creates uncertainty and leaves staff without a suitable space for urgent or important meetings. Separate informal bookings also make it difficult to identify the person responsible for a reservation. These problems are described in the supplied project presentation.

1.2 Business Objective

Provide one shared reservation record for advance and Walk-in bookings, show room facilities and available time slots, and validate availability before confirmation. Enable administrators to identify bookers and cancel reservations when necessary. Availability represents recorded reservations; physical occupancy detection is outside this release.

1.3 Product Vision

For employees who need meeting rooms and administrators who manage them, MEETSPACE provides room discovery, shared availability and confirmed reservations through a member website and a separate Walk-in kiosk so that staff can select suitable rooms and reduce conflicting bookings.

1.4 Business Scope

Release 1 includes public room browsing, account registration and authentication, member reservations, Walk-in reservations with a name or employee identifier, personal booking history, cancellation, notifications, room administration and user role administration. Both channels share a Spring Boot service and H2 database. The current booking flow confirms valid requests automatically and does not require a meeting title, participant count or weekly recurrence input.

Approval workflows, recurring bookings, booking modification, physical occupancy sensors, employee identity verification, payments and email or SMS delivery are not part of the active Release 1 user flow. Some legacy database fields and server actions remain for compatibility and do not establish an enabled user feature.

1.5 Business Workflow

Member workflow: sign in, browse or select a room, inspect equipment and availability, choose a date and start and end time, review and submit. The server validates the room and time and checks shared availability. A valid request is saved as confirmed with an event record. The member views the result in My Bookings and may cancel an eligible reservation. Invalid or conflicting requests display an error without creating a reservation.

Walk-in workflow: open the separate kiosk route, select a room and time, enter a booker name or employee card identifier, review and submit. The server applies the same availability rules and records the booking under the kiosk system account with the supplied booker identity. An administrator can inspect and cancel the reservation.

Administrator workflow: sign in, open booking management, inspect the booker and source, and cancel an active reservation if required. Room and user management are restricted to administrators.

Business Workflow Diagram

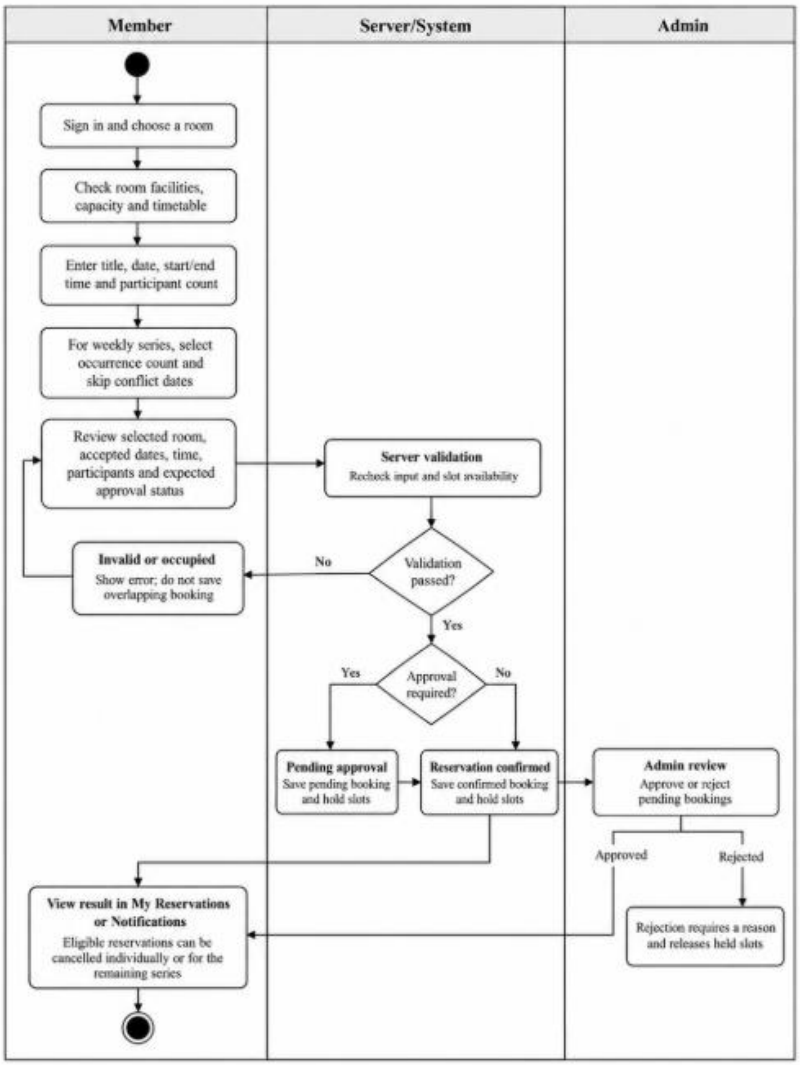


Figure 1 Business Workflow

The reference model includes approval and recurring steps. Release 1 confirms valid bookings automatically and creates single reservations.

1.6 Definitions Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification
FR / NFR	Functional requirement / non functional requirement
Member	Authenticated user of the main website
Visitor	Unauthenticated room and schedule viewer
Walk-in	Separate kiosk reservation channel without personal account login
Confirmed	Reservation that holds a room time slot
Cancelled	Reservation released by its owner or an administrator
Pending / Rejected	Legacy approval states retained in the model
PK / FK / UQ	Primary key / foreign key / unique constraint
Availability	Free or occupied time according to stored active reservations

2 Overall Description

2.1 Product Perspective

MEETSPACE is a stand-alone web application. A browser interface communicates with Spring Boot REST endpoints and stores room, account, reservation and event data in H2. The main website and #walkin route are separate interfaces over the same data, rather than independent reservation stores. Room images may be served through application image storage. Server date and time validation uses Asia/Bangkok.

The project has no demonstrated integration with an employee directory or physical room sensor. Entering an employee identifier on the kiosk records a declaration of identity rather than authenticating that employee. The current prototype must be evaluated before production capacity or availability commitments are made.

2.2 User Classes and Characteristics

User class	**Characteristics and permitted actions**
Visitor	Occasional user; views rooms, facilities and anonymous occupied slots.
Member	Primary user; registers, signs in, reserves rooms and manages own bookings.
Walk-in booker	Uses the kiosk with minimal steps and enters a name or employee identifier; does not obtain member or administrator access.
Administrator	Authorized operator; manages rooms and accounts, sees booker details and cancels reservations.

3 Software Specific Requirements

3.1 Overall Functional Requirements

ID	Function	User class
FR-01	Account registration	Visitor
FR-02	Login and logout	Member and Administrator
FR-03	Browse rooms and facilities	All users
FR-04	View room availability	All users
FR-05	Create member reservation	Member and Administrator
FR-06	Create Walk-in reservation	Walk-in booker
FR-07	Validate reservations	System for both booking channels
FR-08	View personal bookings	Member and Administrator
FR-09	Cancel reservation	Owner or Administrator
FR-10	Manage room information	Administrator
FR-11	Manage users and roles	Administrator
FR-12	View and manage all bookings	Administrator
FR-13	Record events and notifications	System and signed in user
FR-14	Change password	Authenticated user

Use Case Diagram

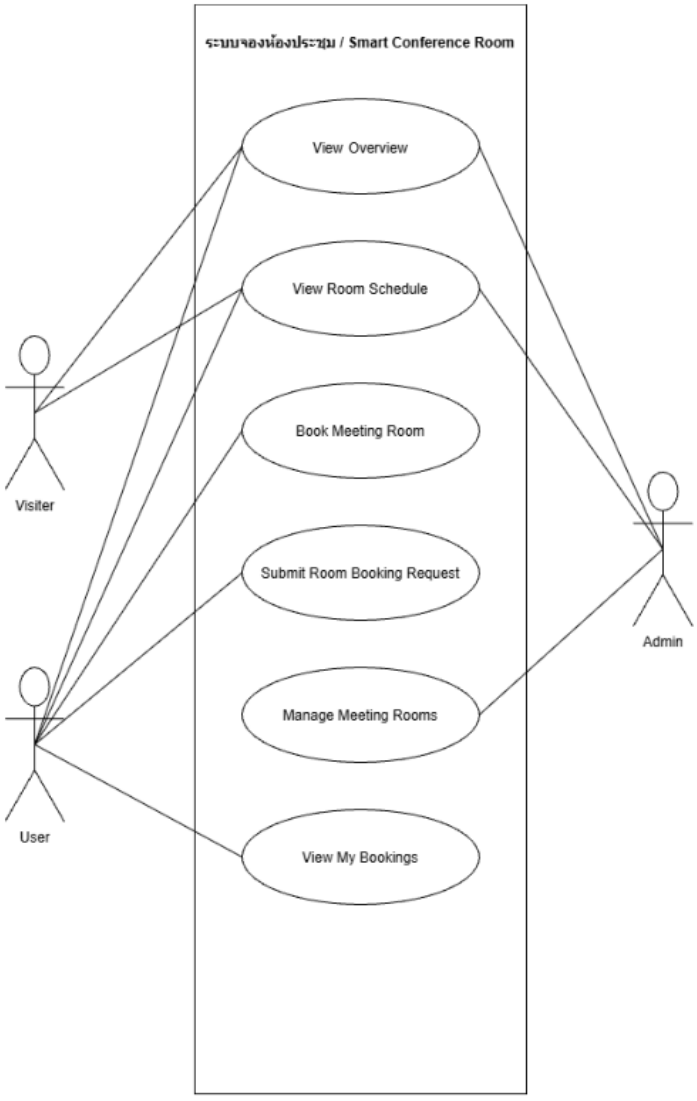


Figure 2 Use Case Diagram

3.2 Functional Requirements

FR 01 Account registration

Requirement ID: FR-01 | Priority: High

Description and output: Create a member account and start a session. Reject blank or overlong names, malformed or duplicate email, and passwords outside 8 to 12 characters.

User class: Visitor. Input: Name, email and password.

Workflow: Enter account details; validate; hash password with BCrypt; save role user; return the member session.

Precondition: The application and database are available; authenticated actions require a valid session.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

FR 02 Login and logout

Requirement ID: FR-02 | Priority: High

Description and output: Authenticate valid credentials and enforce the account role. Reject invalid credentials without starting an authenticated session.

User class: Member and Administrator. Input: Email and password or logout request.

Workflow: Submit credentials; verify password hash; establish session; logout invalidates the session.

Precondition: The application and database are available; authenticated actions require a valid session.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

FR 03 Browse rooms and facilities

Requirement ID: FR-03 | Priority: High

Description and output: Display room name, location, capacity, amenities, image and active state.

User class: All users. Input: Room list or selected room.

Workflow: Load room data; browse or filter in the interface; open selected room details.

Precondition: The application and database are available; authenticated actions require a valid session.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

FR 04 View room availability

Requirement ID: FR-04 | Priority: High

Description and output: Display occupied slots without exposing another booker identity to visitors or ordinary members.

User class: All users. Input: Room and date.

Workflow: Load active reservations; show time slots; refresh data before submitting a reservation.

Precondition: The application and database are available; authenticated actions require a valid session.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

FR 05 Create member reservation

Requirement ID: FR-05 | Priority: High

Description and output: Save a valid request as confirmed. No meeting title is requested from the user.

User class: Member and Administrator. Input: Room, date, start and end time, request key.

Workflow: Select an active room; choose time; review; validate on server; check overlap; save reservation and confirmation event.

Precondition: The application and database are available; authenticated actions require a valid session.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

FR 06 Create Walk-in reservation

Requirement ID: FR-06 | Priority: High

Description and output: Save a confirmed reservation in the shared store and retain a booker identity of 1 to 100 characters.

User class: Walk-in booker. Input: Room, date, time and booker name or employee identifier.

Workflow: Open #walkin; select room and time; enter identity; review; submit to Walk-in endpoint; validate and confirm.

Precondition: The application and database are available; authenticated actions require a valid session.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

FR 07 Validate reservations

Requirement ID: FR-07 | Priority: High

Description and output: Reject past, invalid, conflicting, inactive room or duplicate requests without a new booking.

User class: System for both booking channels. Input: Proposed room, date and interval.

Workflow: Check date no later than one year ahead; use 30 minute boundaries between 08:00 and 20:00; require start before end and duration at most four hours; check active reservations.

Precondition: The application and database are available; authenticated actions require a valid session.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

FR 08 View personal bookings

Requirement ID: FR-08 | Priority: High

Description and output: Show the current account bookings with room, date, time and status.

User class: Member and Administrator. Input: Authenticated account.

Workflow: Authenticate; retrieve own complete records; display current and historical reservations.

Precondition: The application and database are available; authenticated actions require a valid session.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

FR 09 Cancel reservation

Requirement ID: FR-09 | Priority: High

Description and output: Set an active reservation to cancelled, release its slot and record an event. Reject access by another ordinary member.

User class: Owner or Administrator. Input: Reservation ID.

Workflow: Select reservation; confirm cancellation; validate authorization and current state; update status and timestamp.

Precondition: The application and database are available; authenticated actions require a valid session.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

FR 10 Manage room information

Requirement ID: FR-10 | Priority: High

Description and output: Create or update rooms. Reject invalid values and changes affecting active future reservations.

User class: Administrator. Input: Room name and Thai name, locations, capacity, amenities, image and active state.

Workflow: Open administration; enter room details; validate required fields and capacity 1 to 1000; save. Room deactivation replaces destructive deletion in the current flow.

Precondition: An administrator session is required.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

FR 11 Manage users and roles

Requirement ID: FR-11 | Priority: High

Description and output: Create accounts and change permitted user roles. Protect the active administrator and workspace administrator account from prohibited role changes.

User class: Administrator. Input: Name, email, password, role and target account.

Workflow: Verify administrator session; validate input and previous role when supplied; create account or update role.

Precondition: An administrator session is required.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

FR 12 View and manage all bookings

Requirement ID: FR-12 | Priority: High

Description and output: Display member or Walk-in source and booker identity; permit cancellation.

User class: Administrator. Input: Booking list and selected reservation.

Workflow: Open booking management; load full reservation details; inspect source and booker; select cancellation where appropriate.

Precondition: An administrator session is required.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

FR 13 Record events and notifications

Requirement ID: FR-13 | Priority: High

Description and output: Store confirmation and cancellation events and show the signed in user notifications.

User class: System and signed in user. Input: Booking ID, recipient, event type, time and actor.

Workflow: Create event with reservation changes; retrieve events for the current account in descending event time.

Precondition: The application and database are available; authenticated actions require a valid session.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

FR 14 Change password

Requirement ID: FR-14 | Priority: High

Description and output: Verify current credentials, store a BCrypt hash for an 8 to 12 character new password, then invalidate the session.

User class: Authenticated user. Input: Current password and new password.

Workflow: Open account controls; submit passwords; validate; update hash; require login again.

Precondition: The application and database are available; authenticated actions require a valid session.

Acceptance: A valid submission produces the stated output; invalid input or unauthorized requests produce an error and do not perform the requested change.

3.3 Non Functional Requirements

The following are acceptance targets for verification, not claims of completed performance or production testing.

ID	Requirement and verification
NFR-01	Passwords must never be returned in public API responses or stored as plaintext; inspect database hashes and account responses.
NFR-02	Direct requests without a required session must return 401; non administrators must receive 403 for administrator operations.
NFR-03	Public availability responses must exclude account names, email addresses and kiosk booker identities; inspect both public data endpoints.
NFR-04	Main and kiosk booking attempts for overlapping intervals must not both obtain the same room slot; run a simultaneous submission test before production deployment.
NFR-05	Reservation validation must use Asia/Bangkok consistently at date and time boundaries; test same day, midnight and past time cases.
NFR-06	Proposed performance target: 95 percent of room and booking requests complete within 2 seconds with 20 concurrent users in a documented test environment.
NFR-07	At 1366 by 768 desktop and 390 by 844 mobile viewports, primary forms must remain readable and usable without clipped controls; inspect both channels.
NFR-08	Confirmed must be labeled in text and shown in green; failed or cancelled outcomes must be labeled and shown in red so color is not the only cue.
NFR-09	A failed validation must leave no partial booking or event; verify row counts and transaction rollback using invalid requests.

4 User Interface Requirements

4.1 Navigation Structure and Sitemap

Main website: overview → room browser → room details and availability → booking review → result. Account entry → registration or login → My Bookings and notifications. Administrator entry → room management, booking management and user management. Walk-in: separate #walkin entry → room and time selection → booker identification → review → result. The main sidebar must not include a Walk-in item.

The supplied sitemap is an earlier design reference. Its edit booking, hard delete and contact pages do not establish implemented Release 1 features. The active flow uses cancellation and a new booking when time changes are required.

Navigation Structure

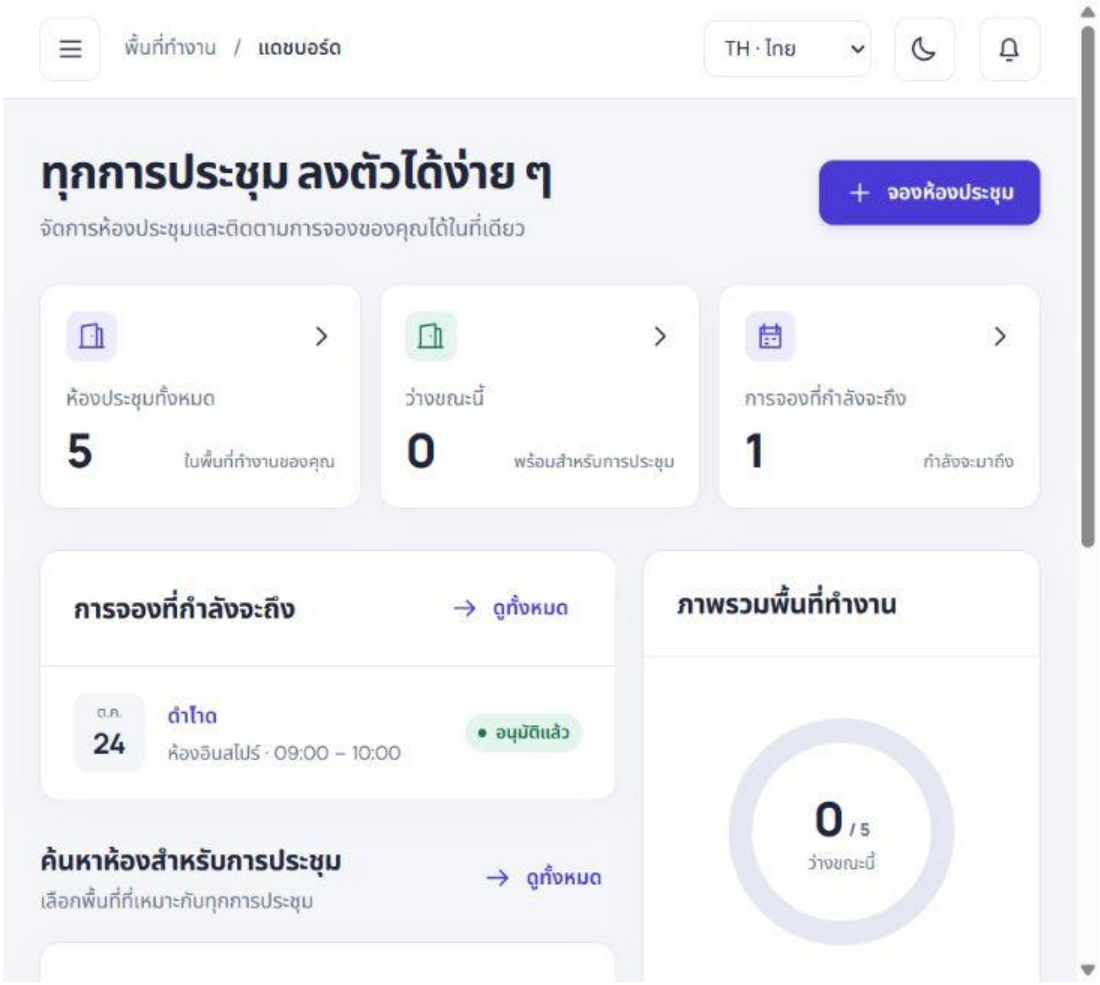


Figure 3 Navigation Structure

4.2 User Interface Specifications

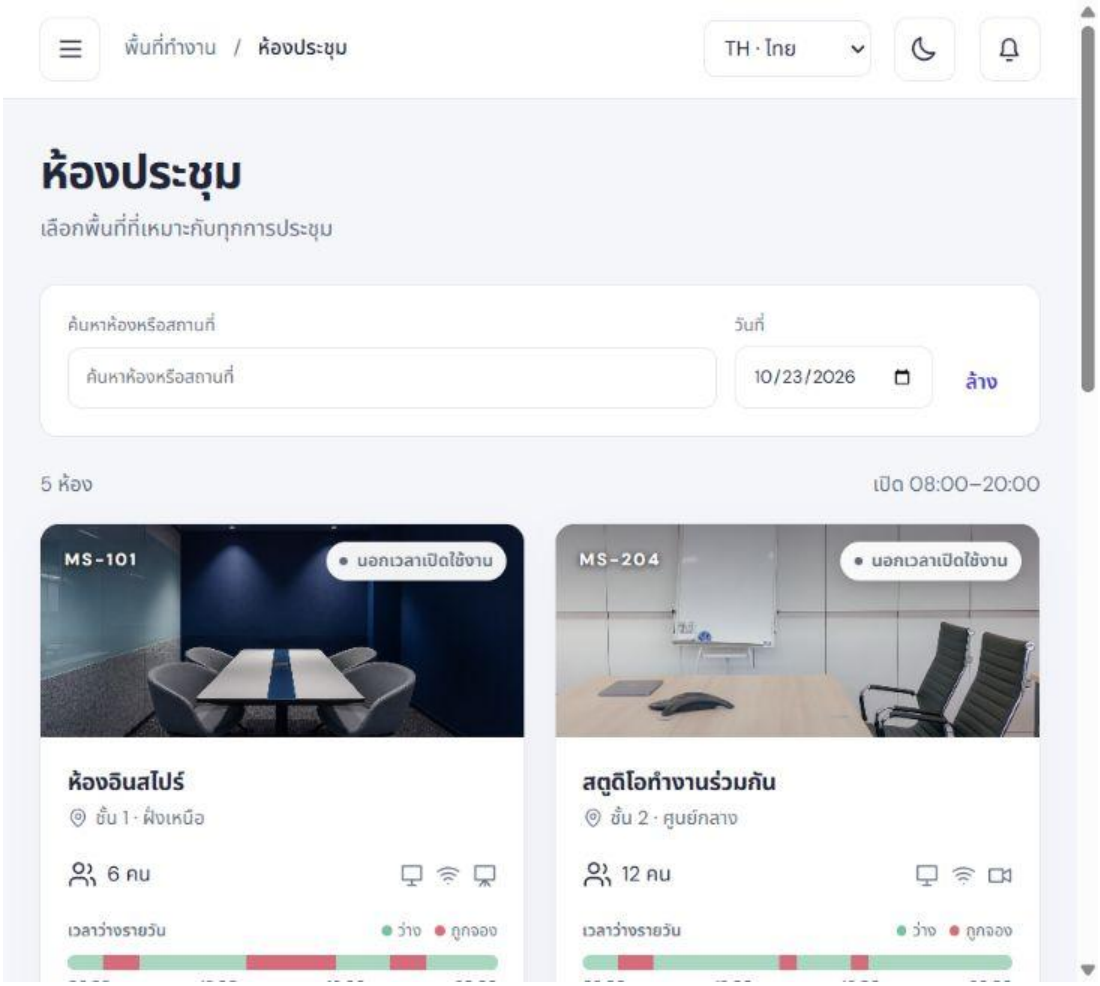
Screen	Required information and behavior	Related FR
Room browser and detail	Room photo, names, location, facilities, capacity and anonymous occupied slots.	FR-03, FR-04
Login and registration	Account fields, validation feedback, clear success or failure message; after sign-in the user returns to the requested page or the Dashboard.	FR-01, FR-02
Main booking form	Room, date, start and end time with review and confirmation. No meeting title field.	FR-05, FR-07
Walk-in kiosk	Separate entry, room/time, required name or employee identifier, review and result; no public identity listing.	FR-06, FR-07
My Bookings	Own reservation details, status, cancellation control and notifications.	FR-08, FR-09, FR-13
Administrator booking management	Booker, member or Walk-in source, room, date, time, status and cancellation.	FR-12
Administrator room management	Validated room attributes and active state with warnings for affected reservations.	FR-10
User management and account controls	Account creation, permitted role changes and password change feedback.	FR-11, FR-14

UI-01 Dashboard



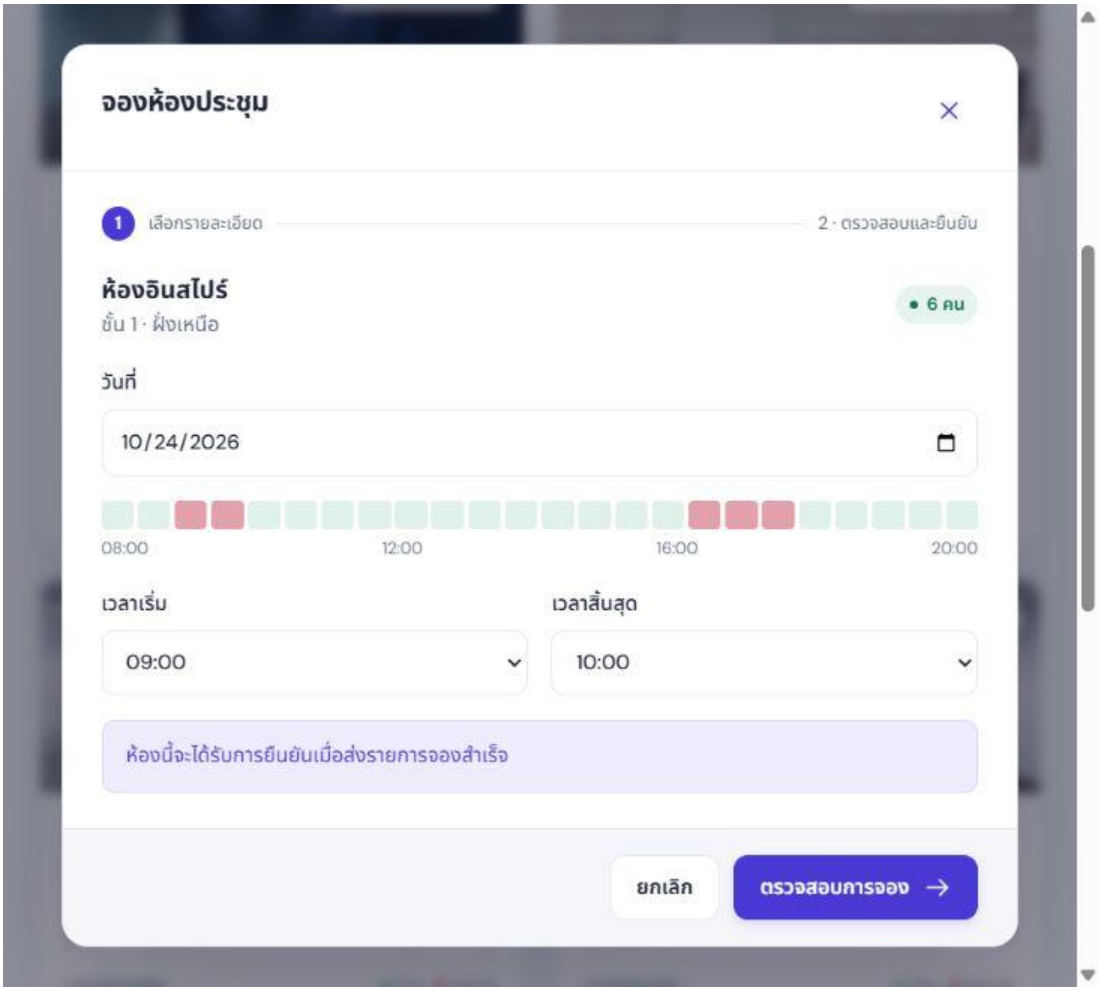
Purpose	Displays booking overview, room availability and upcoming reservations.
Related FR	FR-03, FR-08

UI-02 Room browser and availability



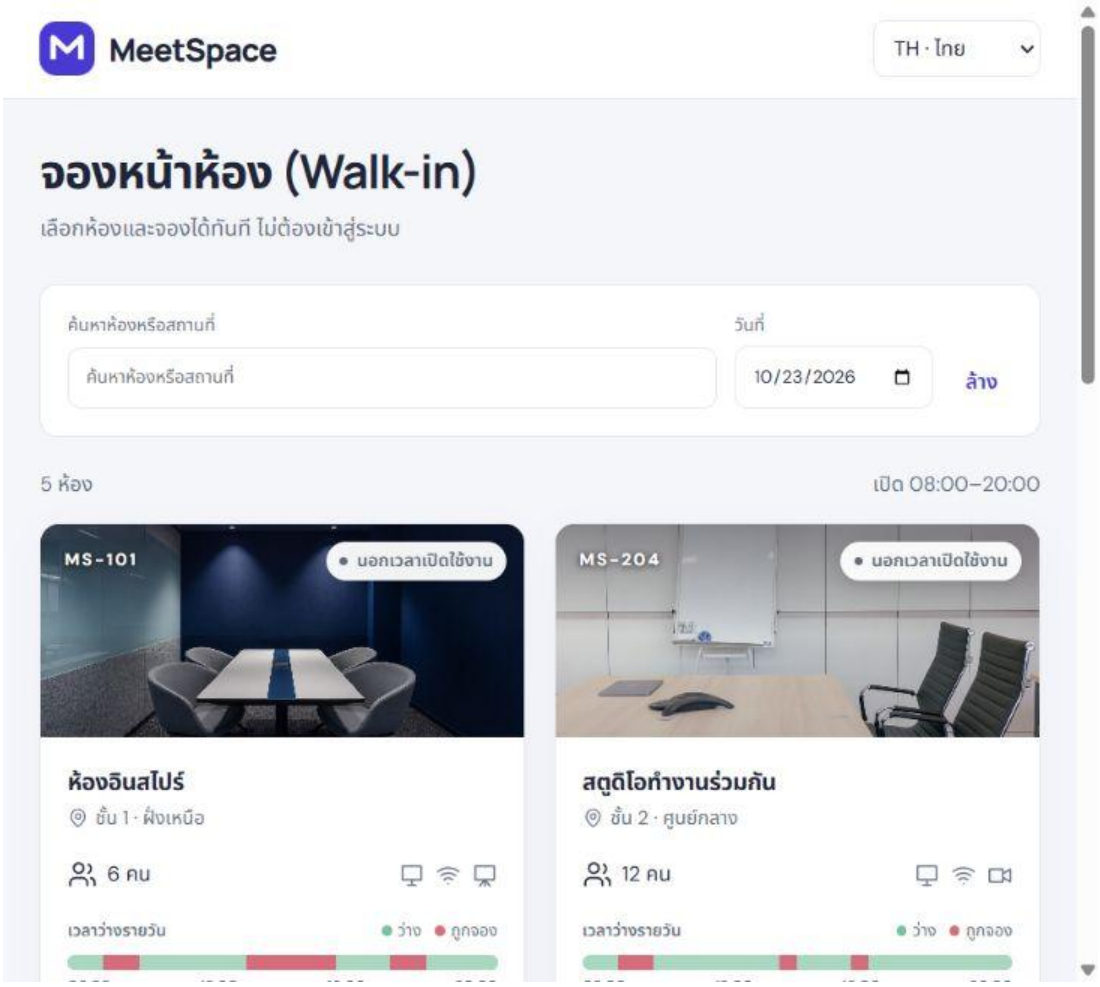
Purpose	Search rooms and inspect location, capacity, facilities and occupied slots.
Related FR	FR-03, FR-04

UI-03 Member booking form

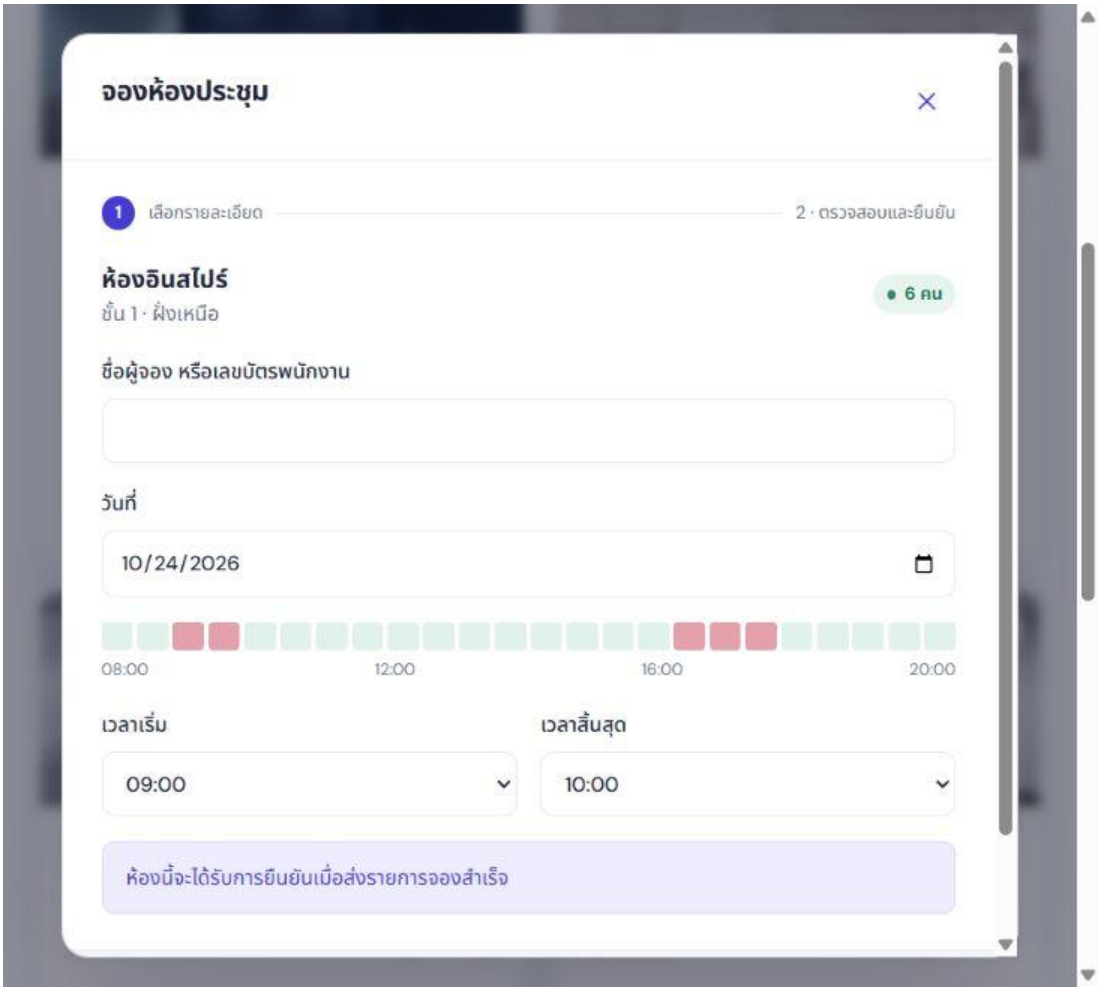


Purpose	Choose date, start and end time, then review before confirming.
Related FR	FR-05, FR-07

UI-04 Walk-in room selection



Purpose	Access the separate kiosk and select a room without member login.
Related FR	FR-06

UI-05 Walk-in booker identification	
	
Purpose	Enter booker name or employee identifier and select a valid time interval.
Related FR	FR-06, FR-07

Entity Relationship Diagram

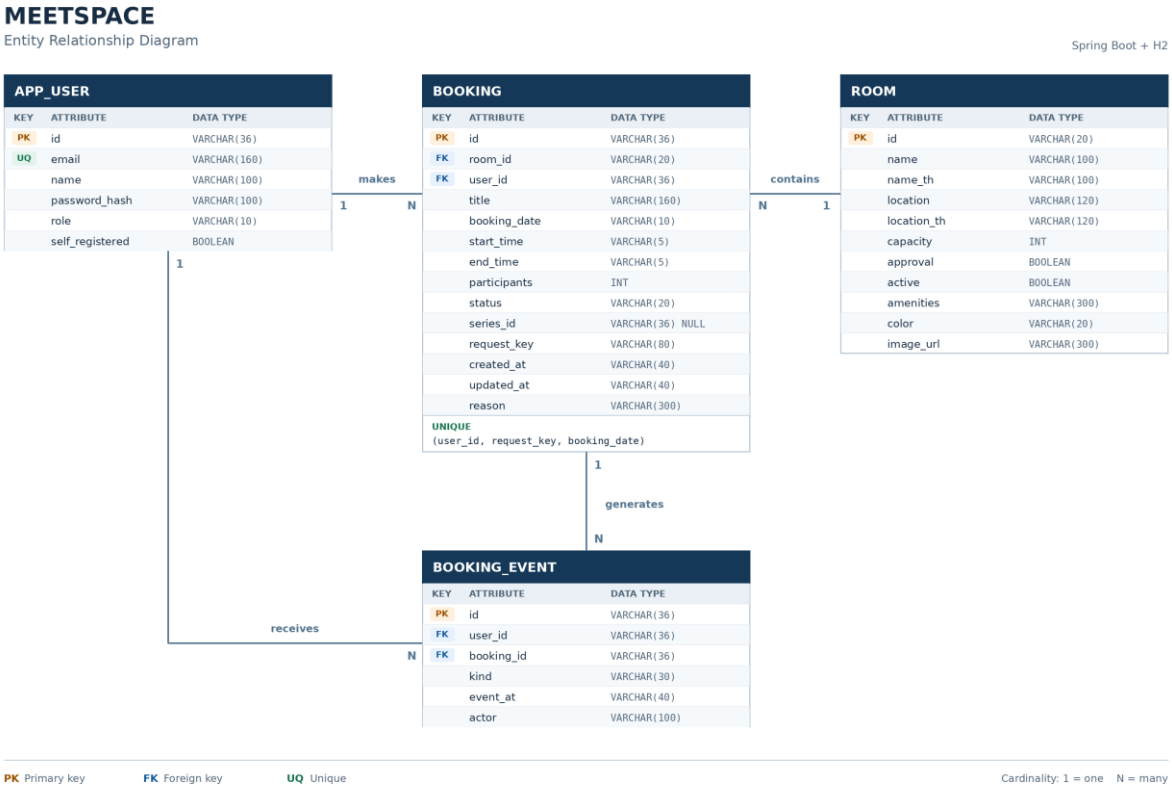


Figure 4 Entity Relationship Diagram

Appendix A Data Dictionary

Types and constraints below follow the current schema. The supplied ERD describes four principal tables; BOOKING.booker and ROOM_IMAGE are additions present in the current source. Dates and times are stored as text in this prototype and must follow the formats stated below.

ROOM_IMAGE

Attribute	Data type	Constraint	Meaning
id	VARCHAR(36)	PRIMARY KEY	Unique record identifier
content	BLOB	NOT NULL	Stored room image bytes

APP_USER

Attribute	Data type	Constraint	Meaning
id	VARCHAR(36)	PRIMARY KEY	Unique record identifier
email	VARCHAR(160)	NOT NULL UNIQUE	Unique account email
name	VARCHAR(100)	NOT NULL	English name or account display name
password_hash	VARCHAR(100)	NOT NULL	BCrypt password hash
role	VARCHAR(10)	NOT NULL	user or admin
self_registered	BOOLEAN	NOT NULL DEFAULT FALSE	Self registration flag

ROOM

Attribute	Data type	Constraint	Meaning
id	VARCHAR(20)	PRIMARY KEY	Unique record identifier
name	VARCHAR(100)	NOT NULL	English name or account display name
name_th	VARCHAR(100)	NOT NULL	Thai room name
location	VARCHAR(120)	NOT NULL	English location
location_th	VARCHAR(120)	NOT NULL	Thai location

Attribute	Data type	Constraint	Meaning
capacity	INT	NOT NULL	Maximum room capacity
approval	BOOLEAN	NOT NULL DEFAULT FALSE	Legacy approval flag; disabled in active release
active	BOOLEAN	NOT NULL DEFAULT TRUE	Room enabled for reservations
amenities	VARCHAR(300)	NOT NULL DEFAULT ''	Comma separated equipment codes
color	VARCHAR(20)	NOT NULL DEFAULT 'indigo'	Interface color identifier
image_url	VARCHAR(300)	NOT NULL DEFAULT ''	Room image reference

BOOKING

Attribute	Data type	Constraint	Meaning
id	VARCHAR(36)	PRIMARY KEY	Unique record identifier
room_id	VARCHAR(20)	NOT NULL	FK to ROOM.id
user_id	VARCHAR(36)	NOT NULL	FK to APP_USER.id; kiosk uses walk-in account
title	VARCHAR(160)	NOT NULL	Internal reservation label; not a user input
booking_date	VARCHAR(10)	NOT NULL	Date in YYYY-MM-DD format
start_time	VARCHAR(5)	NOT NULL	Start in HH:mm format
end_time	VARCHAR(5)	NOT NULL	End in HH:mm format
participants	INT	NOT NULL	Compatibility field; active flow uses 1
status	VARCHAR(20)	NOT NULL	confirmed or cancelled; pending and rejected retained
series_id	VARCHAR(36)	Nullable	Nullable legacy recurring series identifier
request_key	VARCHAR(80)	NOT NULL	Request identifier for duplicate detection
created_at	VARCHAR(40)	NOT NULL	Creation timestamp
updated_at	VARCHAR(40)	NOT NULL	Last update timestamp

Attribute	Data type	Constraint	Meaning
reason	VARCHAR(300)	NOT NULL DEFAULT ''	Legacy rejection reason
booker	VARCHAR(100)	NOT NULL default empty	Declared Walk-in identity or member name

BOOKING_EVENT

Attribute	Data type	Constraint	Meaning
id	VARCHAR(36)	PRIMARY KEY	Unique record identifier
user_id	VARCHAR(36)	NOT NULL	FK to APP_USER.id; kiosk uses walk-in account
booking_id	VARCHAR(36)	NOT NULL	FK to BOOKING.id
kind	VARCHAR(30)	NOT NULL	Event type
event_at	VARCHAR(40)	NOT NULL	Event timestamp
actor	VARCHAR(100)	NOT NULL	Actor display name

Relationships: one APP_USER has many BOOKING records; one ROOM has many BOOKING records; one BOOKING has many BOOKING_EVENT records; one APP_USER receives many BOOKING_EVENT records. BOOKING enforces UNIQUE(user_id, request_key, booking_date). The image table has no declared foreign key to ROOM in the supplied schema.